to computational constraints, The AI Scientist did not always collect additional results; however, in these cases, it would sometimes hallucinate an entire ablations table. We resolved this by instructing The AI Scientist explicitly to only include results it directly observed. Furthermore, it frequently hallucinates facts we do not provide, such as the hardware used.
- More generally, we do not recommend taking the scientific content of this version of The AI Scientist at face value. Instead, we advise treating generated papers as hints of promising ideas for practitioners to follow up on. Nonetheless, we expect the trustworthiness of The AI Scientist to increase dramatically in the coming years in tandem with improvements to foundation models. We share this paper and code primarily to show what is currently possible and hint at what is likely to be possible soon.

Safe Code Execution. The current implementation of The AI Scientist has minimal direct sandboxing in the code, leading to several unexpected and sometimes undesirable outcomes if not appropriately guarded against. For example, in one run, The AI Scientist wrote code in the experiment file that initiated a system call to relaunch itself, causing an uncontrolled increase in Python processes and eventually necessitating manual intervention. In another run, The AI Scientist edited the code to save a checkpoint for every update step, which took up nearly a terabyte of storage. In some cases, when The AI Scientist's experiments exceeded our imposed time limits, it attempted to edit the code to extend the time limit arbitrarily instead of trying to shorten the runtime. While creative, the act of bypassing the experimenter's imposed constraints has potential implications for AI safety (Lehman et al., 2020). Moreover, The AI Scientist occasionally imported unfamiliar Python libraries, further exacerbating safety concerns. We recommend strict sandboxing when running The AI Scientist, such as containerization, restricted internet access (except for Semantic Scholar), and limitations on storage usage.

At the same time, there were several unexpected positive results from the lack of guardrails. For example, we had forgotten to create the output results directory in the grokking template in our experiments. Each successful run from The AI Scientist that outputted a paper automatically caught this error when it occurred and fixed it. Furthermore, we found that The AI Scientist would occasionally include results and plots that we found surprising, differing significantly from the provided templates. We describe some of these novel algorithm-specific visualizations in Section 6.1.

Broader Impact and Ethical Considerations. While The AI Scientist has the potential to be a valuable tool for researchers, it also carries significant risks of misuse. The ability to automatically generate and submit papers to academic venues could greatly increase the workload for reviewers, potentially overwhelming the peer review process and compromising scientific quality control. Similar concerns have been raised about generative AI in other fields, such as its impact on the arts (Epstein et al., 2023). Furthermore, if the Automated Reviewer tool was widely adopted by reviewers, it could diminish the quality of reviews and introduce undesirable biases into the evaluation of papers. Because of this, we believe that papers or reviews that are substantially AI-generated must be marked as such for full transparency.

As with most previous technological advances, The AI Scientist has the potential to be used in unethical ways. For example, it could be explicitly deployed to conduct unethical research, or even lead to unintended harm if The AI Scientist conducts unsafe research. Concretely, if it were encouraged to find novel, interesting biological materials and given access to "cloud labs" (Arnold, 2022) where robots perform wet lab biology experiments, it could (without its overseer's intent) create new, dangerous viruses or poisons that harm people before we can intervene. Even in computers, if tasked to create new, interesting, functional software, it could create dangerous malware. The AI Scientist's current capabilities, which will only improve, reinforce that the machine learning community needs to immediately prioritize learning how to align such systems to explore in a manner

that is safe and consistent with our values.

9. Discussion

In this paper, we introduced THE AI SCIENTIST, the first framework designed to fully automate the scientific discovery process, and, as a first demonstration of its capabilities, applied it to machine learning itself. This end-to-end system leverages LLMs to autonomously generate research ideas, implement and execute experiments, search for related works, and produce comprehensive research papers. By integrating stages of ideation, experimentation, and iterative refinement, THE AI SCIENTIST aims to replicate the human scientific process in an automated and scalable manner.

Why does writing papers matter? Given our overarching goal to automate scientific discovery, why are we also motivated to have THE AI SCIENTIST write papers, like human scientists? For example, previous AI-enabled systems such as FunSearch (Romera-Paredes et al., 2024) and GNoME (Pyzer-Knapp et al., 2022) also conduct impressive scientific discovery in restricted domains, but they do not write papers.

There are several reasons why we believe it is fundamentally important for THE AI SCIENTIST to write scientific papers to communicate its discoveries. First, writing papers offers a highly interpretable method for humans to benefit from what has been learned. Second, reviewing written papers within the framework of existing machine learning conferences enables us to standardize evaluation. Third, the scientific paper has been the primary medium for disseminating research findings since the dawn of modern science. Since a paper can use natural language, and include plots and code, it can flexibly describe any type of scientific study and discovery. Almost any other conceivable format is locked into a certain kind of data or type of science. Until a superior alternative emerges (or possibly invented by AI), we believe that training THE AI SCIENTIST to produce scientific papers is essential for its integration into the broader scientific community.

Costs. Our framework is remarkably versatile and effectively conducts research across various subfields of machine learning, including transformer-based language modeling, neural network learning dynamics, and diffusion modeling. The cost-effectiveness of the system, producing papers with potential conference relevance at an approximate cost of $15 per paper, highlights its ability to democratize research (increase its accessibility) and accelerate scientific progress. Preliminary qualitative analysis, for example in Section 5, suggests that the generated papers can be broadly informative and novel, or at least contain ideas worthy of future study.

The actual compute we allocated for THE AI SCIENTIST to conduct its experiments in this work is also incredibly light by today's standards. Notably, our experiments generating hundreds of papers were largely run only using a single 8×NVIDIA H100 node over the course of a week. Massively scaling the search and filtering would likely result in significantly higher-quality papers.

In this project, the bulk of the cost for running THE AI SCIENTIST is associated with the LLM API costs for coding and paper writing. In contrast, the costs associated with running the LLM reviewer, as well as the computational expenses for conducting experiments, are negligible due to the constraints we've imposed to keep overall costs down. However, this cost breakdown may change in the future if THE AI SCIENTIST is applied to other scientific fields or used for larger-scale computational experiments.

Open vs. Closed Models. To quantitatively evaluate and improve the generated papers, we first created and validated an Automated Paper Reviewer. We show that, although there is significant room for improvement, LLMs are capable of producing reasonably accurate reviews, achieving results comparable to humans across various metrics. Applying this evaluator to the papers generated by THE AI SCIENTIST enables us to scale the evaluation of our papers beyond manual inspection.

We find that Sonnet 3.5 consistently produces the best papers, with a few of them even achieving a score that exceeds the threshold for acceptance at a standard machine learning conference from the Automated Paper Reviewer.

However, there is no fundamental reason to expect a single model like Sonnet 3.5 to maintain its lead. We anticipate that all frontier LLMs, including open models, will continue to improve. The competition among LLMs has led to their commoditization and increased capabilities. Therefore, our work aims to be model-agnostic regarding the foundation model provider. In this project, we studied various proprietary LLMs, including GPT-4o and Sonnet, but also explored using open models like DeepSeek and Llama-3. We found that open models offer significant benefits, such as lower costs, guaranteed availability, greater transparency, and flexibility, although slightly worse quality. In the future, we aim to use our proposed discovery process to produce self-improving AI in a closed-loop system using open models.

Future Directions. Direct enhancements to The AI Scientist could include integrating vision capabilities for better plot and figure handling, incorporating human feedback and interaction to refine the AI's outputs, and enabling The AI Scientist to automatically expand the scope of its experiments by pulling in new data and models from the internet, provided this can be done safely. Additionally, The AI Scientist could follow up on its best ideas or even perform research directly on *its own code* in a self-referential manner. Indeed, significant portions of the code for this project were written by Aider. Expanding the framework to other scientific domains could further amplify its impact, paving the way for a new era of automated scientific discovery. For example, by integrating these technologies with cloud robotics and automation in physical lab spaces (Arnold, 2022; Kehoe et al., 2015; Sparkes et al., 2010; Zucchelli et al., 2021) provided it can be done safely, The AI Scientist could perform experiments for biology, chemistry, and material sciences.

Crucially, future work should address the reliability and hallucination concerns, potentially through a more in-depth automatic verification of the reported results. This could be done by directly linking code and experiments, or by seeing if an automated verifier can independently reproduce the results.

Conclusion. The introduction of The AI Scientist marks a significant step towards realizing the full potential of AI in scientific research. By automating the discovery process and incorporating an AI-driven review system, we open the door to endless possibilities for innovation and problem-solving in the most challenging areas of science and technology. Ultimately, we envision a fully AI-driven scientific ecosystem including not only AI-driven researchers but also reviewers, area chairs, and entire conferences. However, we do not believe the role of a human scientist will be diminished. We expect the role of scientists will change as we adapt to new technology, and they will be empowered to tackle more ambitious goals. For instance, researchers often have more ideas than they have time to pursue, what if The AI Scientist could take the first explorations on all of them?

While the current iteration of The AI Scientist demonstrates a strong ability to innovate on top of well-established ideas, such as Diffusion Modeling or Transformers, it is an open question whether such systems can ultimately propose genuinely paradigm-shifting ideas. Will future versions of The AI Scientist be capable of proposing ideas as impactful as Diffusion Modeling, or come up with the next Transformer architecture? Will machines ultimately be able to invent concepts as fundamental as the artificial neural network, or information theory? We believe The AI Scientist will make a great *companion* to human scientists, but only time will tell to the extent to which the nature of human creativity and our moments of serendipitous innovation (Stanley and Lehman, 2015) can be replicated by an open-ended discovery process conducted by artificial agents.